

Open in app ↗

 Search Write

★ Get unlimited access to the best of Medium for less than \$1/week. [Become a member](#)



PicoCTF — Vault Door 1



EtcSec

[Follow](#)

3 min read · Sep 30, 2022



Platform: PicoCTF

Challenge: vault-door-1

Category: Reverse Engineering

Point Value: 100

Attack System:

Kali Linux / Current

Before beginning Vault Door 1, I ran through the Vault Door Training challenge.

vault-door-training

Tags: picoCTF 2019 Reverse Engineering

AUTHOR: MARK E. HAASE

Description

Your mission is to enter Dr. Evil's laboratory and retrieve the blueprints for his Doomsday Project. The laboratory is protected by a series of locked vault doors. Each door is controlled by a computer and requires a password to open. Unfortunately, our undercover agents have not been able to obtain the secret passwords for the vault doors, but one of our junior agents obtained the source code for each vault's computer! You will need to read the source code for each level to figure out what the password is for that vault door. As a warmup, we have created a replica vault in our training facility. The source code for the training vault is here: [VaultDoorTraining.java](#)

I downloaded VaultDoorTraining.java and took a peek at the code to see what was there. It appeared to be a simple Java program that took a “password” input (the flag) from the user, and checked to validate that it was correct.

The “password” was clearly visible on line 24.

```
1 import java.util.*;
2
3 class VaultDoorTraining {
4     public static void main(String args[]) {
5         VaultDoorTraining vaultDoor = new VaultDoorTraining();
6         Scanner scanner = new Scanner(System.in);
7         System.out.print("Enter vault password: ");
8         String userInput = scanner.next();
9         String input = userInput.substring("picoCTF{".length(),userInput.length()-1);
10        if (vaultDoor.checkPassword(input)) {
11            System.out.println("Access granted.");
12        } else {
13            System.out.println("Access denied!");
14        }
15    }
16
17    // The password is below. Is it safe to put the password in the source code?
18    // What if somebody stole our source code? Then they would know what our
19    // password is. Hmm... I will think of some ways to improve the security
20    // on the other doors.
21    //
22    // -Minion #9567
23    public boolean checkPassword(String password) {
24        return password.equals("w4rm1ng_Up_w1tH_jAv4_be8d9806f18");
25    }
26 }
```

As proof of functionality, I ran the program and tested it out. The first time I entered the password, the response was “Access denied!”. Oops, I forgot the rest of the flag... The second time I ran the program, the response was “Access granted.”

```
(etcsec)-[picoctf/vault_door_training]
$ java VaultDoorTraining.java
Picked up _JAVA_OPTIONS: -Dawt.useSystemAAFontSettings=on -Dswing.aatext=true
Enter vault password: picoCTF{w4rm1ng_Up_w1tH_jAv4_be8d9806f18}
Access granted.
```

With the flag ‘validated’, I entered it into the challenge solution box in the PicoCTF “Practice Gym”, and was awarded 50 points.

Now that I had a grasp on what was required for these challenges, I moved on to Vault Door 1.

vault-door-1

Tags: picoCTF 2019 Reverse Engineering

AUTHOR: MARK E. HAASE

Description

This vault uses some complicated arrays! I hope you can make sense of it, special agent. The source code for this vault is here: [VaultDoor1.java](#)

I downloaded the VaultDoor1.java file and opened it in a code editor called Geany. As expected, it was a Java program. The structure was the same as in the 'training' challenge, except for the checkPassword() function.



In the checkPassword() function, I could see that an array of 32 characters was being built, but not in a human readable order.

```

public boolean checkPassword(String password) {
    return password.length() == 32 &&
        password.charAt(0) == 'd' &&
        password.charAt(29) == '9' &&
        password.charAt(4) == 'r' &&
        password.charAt(2) == '5' &&
        password.charAt(23) == 'r' &&
        password.charAt(3) == 'c' &&
        password.charAt(17) == '4' &&
        password.charAt(1) == '3' &&
        password.charAt(7) == 'b' &&
        password.charAt(10) == ' ' &&
        password.charAt(5) == '4' &&
}

```

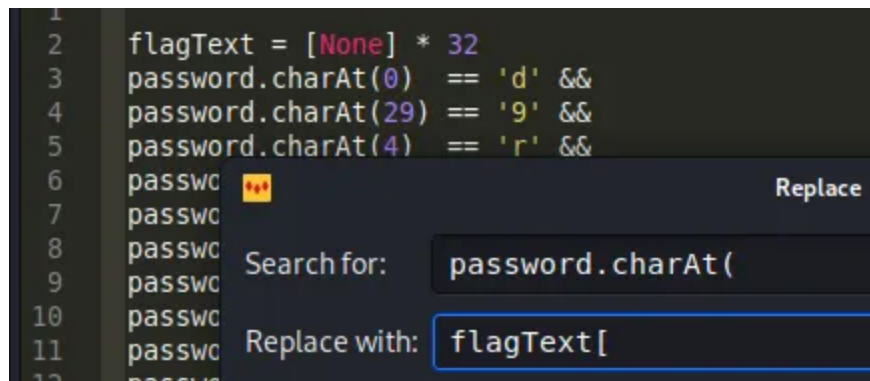
Clearly I needed to find a way to unscramble the letters if I wanted the flag text. I could have just modified the Java code to print the flag, but I instead chose to use Python3 for this. I opened a new file (within Geany) and copied/pasted the checkPassword() function into the new file. I created a new list named 'flagText' with a length of 32. Then, I used find/replace tool in Geany to convert the Java syntax to something that Python3 would agree with.

Required Replacements:

```

password.charAt( --> flagText[
) --> ]
== --> =
&& --> [blank]

```



Replacing password.charAt(with flagText{

```

1  flagText = [None] * 32
2  flagText[0] == 'd' &&
3  flagText[29] == '9' &&
4  flagText[4] == 'r' &&
5  flagText[2] == '5' &&
6  flag
7  flag
8  flag
9  flag
10 flag
11 flag
12 flag
13 flag

```

Search for: ==

Replace with: =

Replace

Replacing == with =

```

1  flagText = [None] * 32
2  flagText[0] = 'd'
3  flagText[29] = '9'
4  flagText[4] = 'r'
5  flagText[2] = '5'
6  flagText[23] = 'r'
7  flagText[3] = 'c'
8  flagText[17] = '4'
9  flagText[1] = '3'
10 flagText[7] = 'b'
11 flagText[10] = ' '
12

```

Code sample after all replacements

With all of the replacements complete, I wrote a little bit of code at the bottom of the program to print “picoCTF{“ and omit the default \n line ending. Once that was on the screen, the loop would step through the flagText list, printing each character, in order on the screen. In the last step, a “}” was printed to finish the flag text allowing for an easy copy/paste flag submission to complete the challenge.

```

35
36 print("picoCTF{",end='')
37 for i in flagText: print(i,end='')
38 print("}\n")
39

```

end=” omits the default \n and allows in-line screen output

I ran the now finished program, and was rewarded with the flag text. This was submitted to complete the challenge, and I was awarded 100 points.

```
(etcsec)-[picocf/vault_door_1]  
$ python3 solve1.py  
picoCTF{[REDACTED]}
```

Hope you enjoyed :)

Happy Hacking!!

[Picoctf](#)[Cybersecurity](#)[Hacking](#)[Walkthrough](#)[Ctf](#)

Written by EtcSec

7 followers · 3 following

Cybersecurity and hacking enthusiast.

[Follow](#)

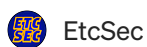
No responses yet



Suzieq

What are your thoughts?

More from EtcSec

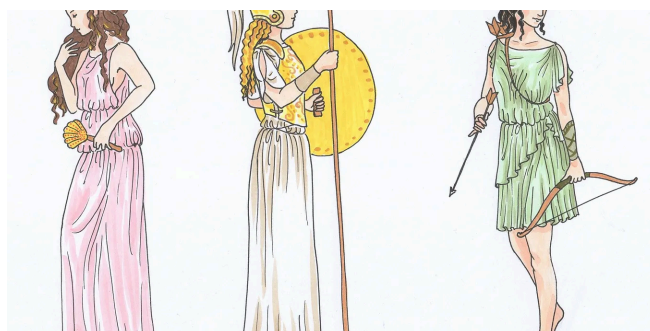


EtcSec

THM—b3dr0ck

Platform: TryHackMe Room: b3dr0ck
Difficulty: Easy Tags: CTF, TLS, socat, socket...

Aug 31, 2022



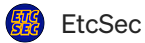
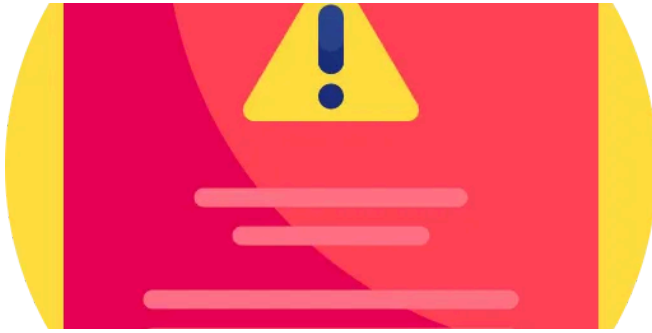
EtcSec

THM—Olympus

Platform: TryHackMe Room: Olympus
Difficulty: Medium Tags: Enumeration, vHost...

Jul 21, 2022



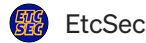


EtcSec

THM—Vulnnet: Endgame

Platform: TryHackMe Room: Vulnnet:
Endgame Difficulty: Medium Tags: Security,...

Sep 17, 2022 🖱 1



EtcSec

THM—Annie

Platform: TryHackMe Room: Annie Difficulty:
Medium Tags: Security, Exploit, boot2root...

Jul 7, 2022 💬 1

[See all from EtcSec](#)

Recommended from Medium

AUTHOR: VENAX

Description

I made a cool website where you can announce whatever you want! Try it out!
Additional details will be available after launching your challenge instance.

This challenge launches an instance on demand.
Its current status is:
`NOT_RUNNING`

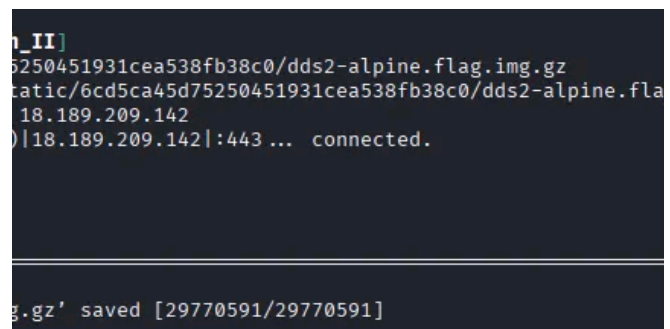
[Launch Instance](#)

Hints ?

1

37,165 users solved

95% Liked



Om Fulsundar



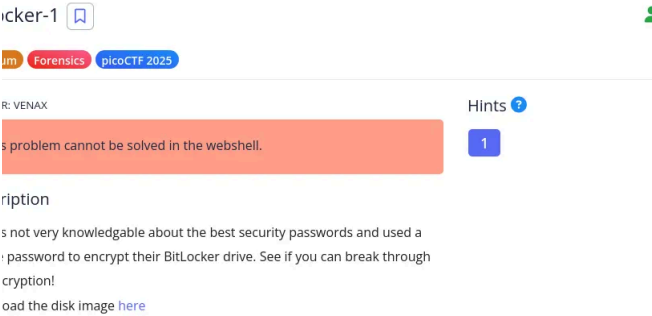
Cyb3rT4ct1c14n-23

SSTI1 & SSTI2—PicoCTF Writeup

These both ctf are based on server side template injection vulnerability.. lets see one...

Nov 19, 2025

1



ColdByte

PicoCTF: Bitlocker-1

This short write-up outlines the approach used to solve the PicoCTF challenge...

Nov 22, 2025



BlueSh3ll

HTB — Interpreter Write-UP | BlueSh3ll

Hello! This is my write-up for Interpreter Medium Linux Machines Release Date 21 Fe...

Step-by-Step Guide to Solving “Disk, disk, sleuth! II”

Hey, disk diviner! This medium-level (200-point) forensics challenge from picoCTF 202...

Oct 23, 2025



Fathijem

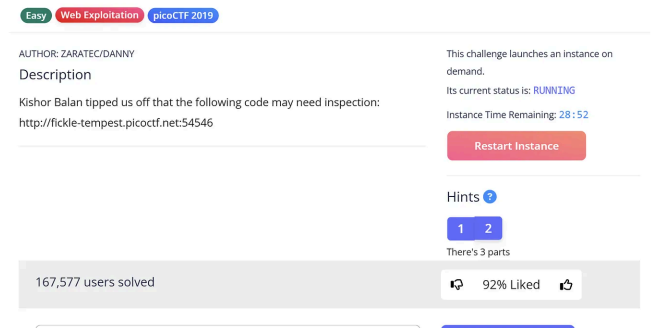
picoMini 2025 CTF Writeup

picoMini CTF 2025 was conducted from October 1–15, 2025, featuring a variety of...

Oct 21, 2025

1

1



Abd Elrahman Reda

Insp3ct0r PicoCTF

after accessing the website i found this

Feb 28  1



3d ago



See more recommendations